

Synthetic Market Relational Representation

Follow-up on Phase 6 profile invariance. This report replaces row retrieval as the main object and asks a more representation-faithful question: are pairwise market relations and whole-snapshot geometry more stable than single-row identity under nuisance variation?

DATE	PROMPTS	SCENARIOS	ANALYSIS
21 March 2026	48 market-only	2 tied-tradeoff families	relational + set-level

MAIN READ

Row retrieval was the wrong object. Pairwise market relations survive nuisance variation far better than single-row identity.

Under the hardest layout-only control, single-row profile identity is nearly gone. But pairwise relation invariance remains strong, with margins roughly 15–20× larger. Whole-snapshot geometry survives too, but more weakly. The best current reading is that the model’s market understanding is more relational than row-identitarian.

ROW IDENTITY	PAIRWISE RELATION	SNAPSHOT GEOMETRY	BEST READ
0.011	0.162	0.00093	Relational
best layout-only margin	best layout-only margin	best layout-only margin	not row retrieval

Why This Report Exists

Phase 6 already established two things:

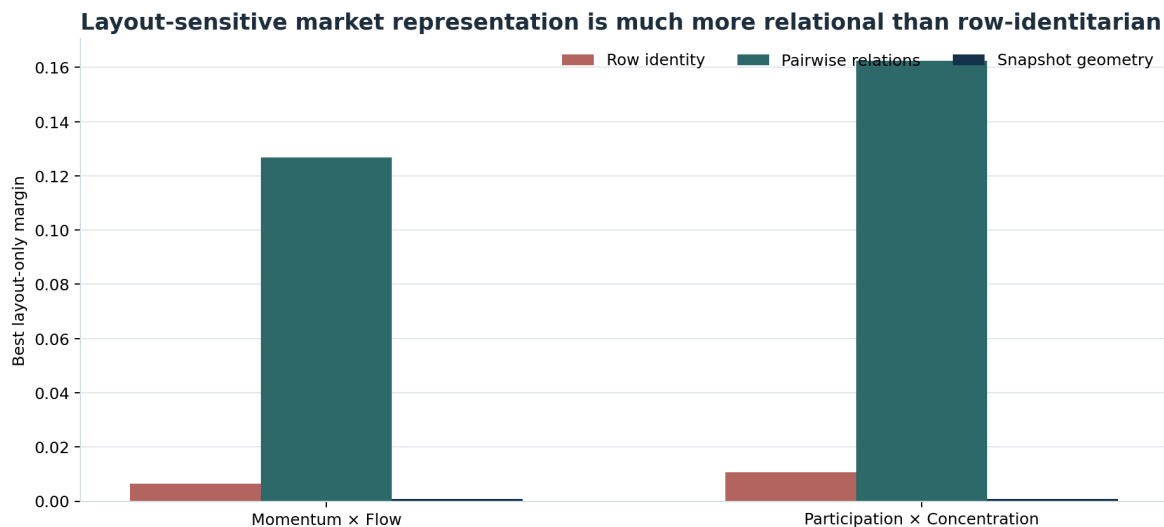
- primitive market factors are explicit in row states
- single-row profile retrieval is fragile under layout changes

That left an obvious methodological question: was the model failing to abstract the market, or were we probing the wrong object?

This report answers that by shifting from **row identity** to two more faithful representation objects:

- **pairwise relations**: how one asset is represented relative to another
- **whole-snapshot geometry**: whether the market as a set retains the same latent shape

The Comparison That Matters

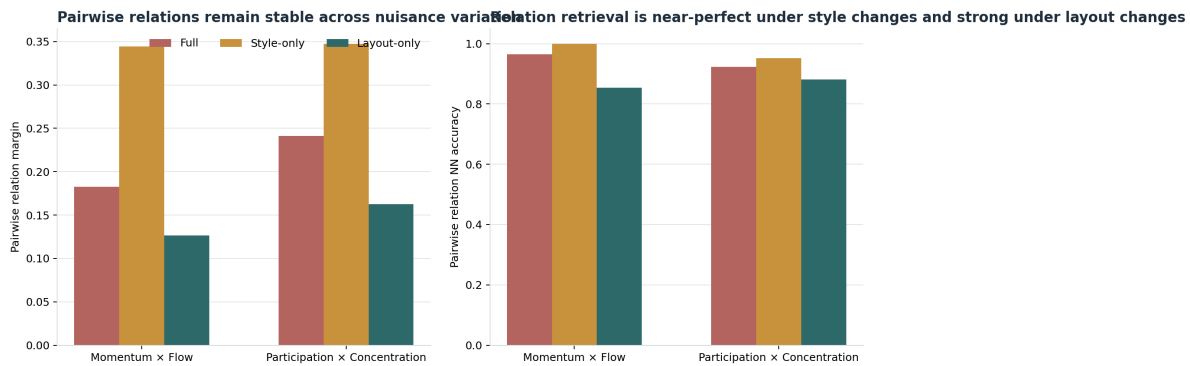


Layout-only margins across three candidate representation objects. Row identity is weak, pairwise relations are much stronger, and whole-snapshot geometry is present but secondary.

Scenario	Object	Margin	NN Acc.	Read
momentum × flow	row identity	0.006	0.667	Barely stable under layout changes.
momentum × flow	pairwise relation	0.127	0.854	Much stronger comparative structure.
momentum × flow	snapshot geometry	0.00085	0.583	Weak but above pure collapse.
participation × concentration	row identity	0.011	0.719	Still weak as a single-row object.
participation × concentration	pairwise relation	0.162	0.882	Strongest current abstraction read.
participation × concentration	snapshot geometry	0.00093	0.667	Set-level structure survives somewhat.

This is the key update. The earlier row-retrieval stress test was useful, but it was not the right scientific object. Once the same latent market is examined in pairwise-relative form, the signal becomes much more robust.

Pairwise Relations Are The Strongest Current Object



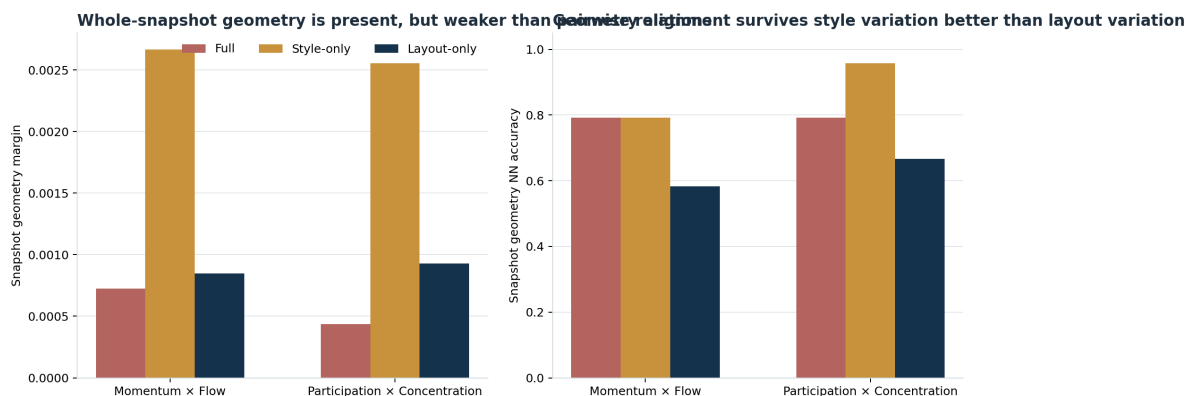
Pairwise relation margins and nearest-neighbor accuracy across nuisance modes. Style changes barely matter. Layout changes reduce the signal, but far less than they did for row identity.

Scenario	Mode	Best Margin	Best NN Acc.
momentum × flow	full	0.182	0.965
momentum × flow	style-only	0.344	1.000
momentum × flow	layout-only	0.127	0.854
participation × concentration	full	0.241	0.924
participation × concentration	style-only	0.348	0.951
participation × concentration	layout-only	0.162	0.882

Interpretation:

- style changes are not the bottleneck
- layout changes still matter, but relations remain strongly recoverable
- participation × concentration remains the stronger family
- the comparative market representation is more stable than the single-row one

Whole-Snapshot Geometry Is Secondary But Real



Whole-snapshot geometry survives nuisance variation more weakly than pairwise relations. Style-only alignment remains respectable; layout-only alignment is modest.

Scenario	Mode	Best Margin	Best NN Acc.
momentum × flow	full	0.00072	0.792
momentum × flow	style-only	0.00267	0.792
momentum × flow	layout-only	0.00085	0.583
participation × concentration	full	0.00044	0.792
participation × concentration	style-only	0.00256	0.958
participation × concentration	layout-only	0.00093	0.667

This is not yet a strong global-manifold result. But it does suggest that some whole-market set structure survives nuisance variation even after row identity itself becomes unreliable.

Interpretation

SUPPORTED NOW

The model's market representation is better described as comparative structure over assets than as stable identity of isolated row embeddings.

NOT SUPPORTED YET

We still do not have evidence for a strongly layout-invariant row-profile representation or a clean global market manifold.

Best current reading:

- primitive factors are explicit in row states
- row retrieval is too brittle to be the main object
- pairwise relations survive nuisance variation well enough to be the best current upstream representation target
- whole-snapshot geometry is worth tracking, but is not yet the strongest claim

Next Step

The right next phase is not more row retrieval.

- Build a harder **relation-first** synthetic dataset with near ties and rank-vs-magnitude controls.
- Hold pairwise latent relations fixed while varying roster composition and distractors.
- Measure whether relation vectors, not row identity, remain stable.
- Then validate the strongest relation family on real DX rows.